Question 072 | Question bank page 7 | Q.1

Write a C# program to demonstrate delegates and events.

Project type: C# Console Application

Steps to perform

1. Open Visual Studio and choose Create a new project.

2. Select Console App (C#), choose a project name and create it.

3. Open Program.cs and replace the sample code with the provided C# code.

4. Build the project; resolve any missing using directives or compilation errors.

5. Press Ctrl+F5, enter the required inputs, and verify the console output.

Closest supplied practical: } } Write the program with Using Delegates and events

Source: Dot_Net_Practicals_1_to_5_.pdf

Provided creation instructions and source files

```
Creation steps: Create the Console App named . Add the listed files at the locations below and enter
the corresponding
code.
Filename: Program.cs    Location: Project root
using System.Text;
using System.Threading.Tasks;
namespace DelegatesEvents
{
    public delegate void NotifyHandler(string message);
    class EventPublisher
    {
        public event NotifyHandler OnDataProcessed;
        public void ProcessData(string data)
        {
            Console.WriteLine($"processing data: {data}");
            OnDataProcessed?.Invoke($"Data '{data}' has been processed!");
        }
    }
    class EventSubscriber
    {
        public void HandleEvent(string message)
        {
            Console.WriteLine($"[Subscriber] Received: {message}");
        }
    }
    delegate int MathOperation(int x, int y);
    class Program
    {
        static int Multiply(int a, int b) => a * b;
        static int Add(int a, int b) => a + b;
        static void Main(string[] args)
        {
            Console.WriteLine("===Delegates Demo===");
            MathOperation op = Multiply;
            Console.WriteLine($"Multiply(4,5)={op(4, 5)}");
            op = Add;
            Console.WriteLine($"Add(4,4)={op(4, 5)}");
            Console.WriteLine("\n===Event Demo===");
            EventPublisher publisher = new EventPublisher();
            EventSubscriber subscriber = new EventSubscriber();
            publisher.ProcessData("Student Record");
            publisher.ProcessData("Salary Data");
            Console.ReadLine();
        }
    }
}
```

Expected result / verification

The application should meet the exact question requirements. Record actual output or screenshots after running; no execution has been claimed in this manual.